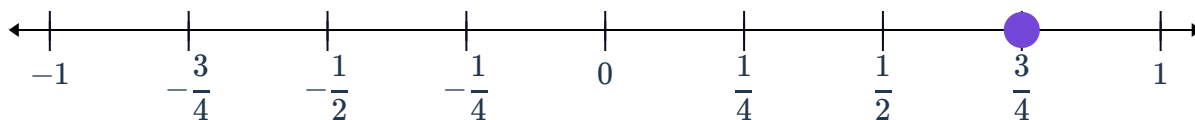


Fractions and the number line

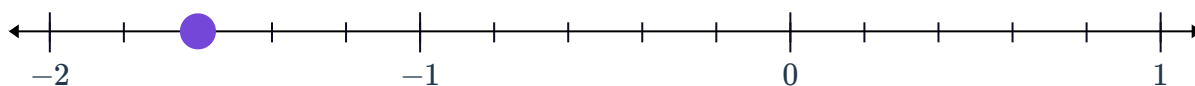


1 State the position of the point plotted on the following number lines as a fraction:

a



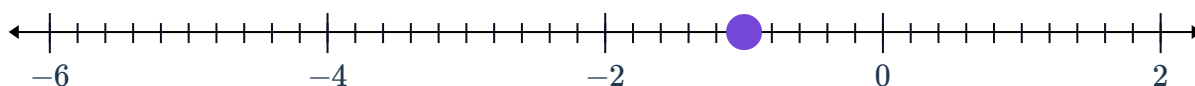
b



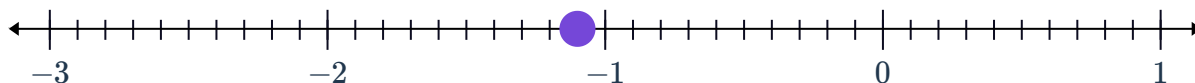
c



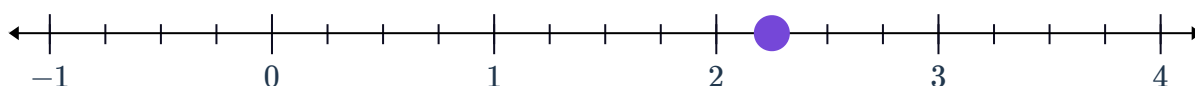
d



e



f



2 For each pair of fractions, state which one is closest to zero:

a $\frac{2}{7}$ and $-\frac{1}{7}$

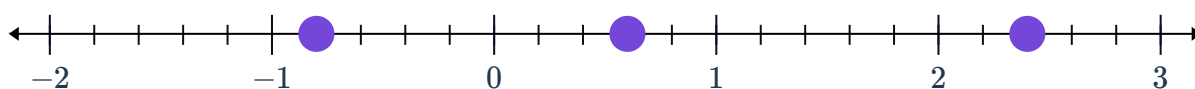
b $-\frac{16}{13}$ and $\frac{7}{13}$

c $\frac{2}{11}$ and $-\frac{4}{11}$

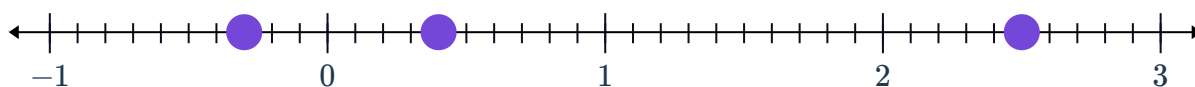
d $-\frac{1}{4}$ and $\frac{3}{4}$

3 State the greatest number plotted on the following number lines:

a



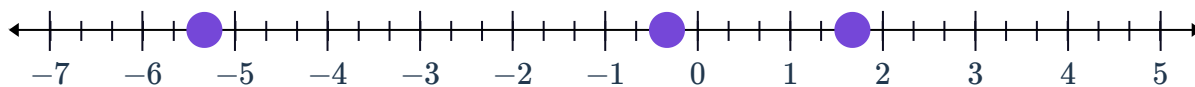
b



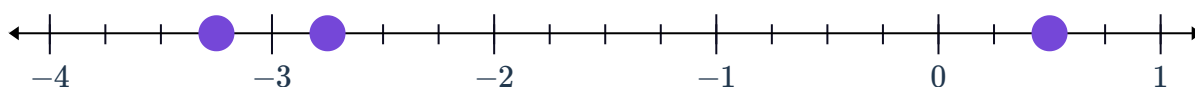
4 State the smallest number plotted on the following number lines:



a



b



Fractions with different denominators

5 State the largest number in each set of fractions:

a $-\frac{5}{3}, -\frac{5}{2}, -\frac{7}{3}$

b $-\frac{1}{2}, \frac{2}{3}, \frac{5}{8}$

c $-\frac{3}{8}, -\frac{9}{16}, -\frac{11}{24}$

d $\frac{27}{10}, \frac{13}{5}, \frac{7}{2}$

6 State the smallest number in each set of fractions:

a $-\frac{1}{5}, -\frac{7}{5}, \frac{8}{3}$

b $-\frac{6}{5}, \frac{21}{10}, -\frac{12}{5}$

c $\frac{9}{4}, \frac{22}{7}, \frac{7}{2}$

d $-\frac{9}{5}, -\frac{15}{8}, -\frac{5}{3}$

7 Arrange the following fractions in ascending order: $-\frac{8}{16}, -\frac{2}{16}, -\frac{3}{16}$

8 Arrange the following fractions in descending order: $-\frac{3}{14}, -\frac{19}{14}, -\frac{6}{14}$

9 For each of the following sets of fractions:

i Rewrite all three fractions with the lowest common denominator.

ii Hence, arrange the fractions in ascending order.

a $-\frac{1}{3}, -\frac{5}{8}, -\frac{7}{24}$

b $-\frac{2}{4}, -\frac{6}{10}, -\frac{7}{40}$



c $-1\frac{5}{7}, -\frac{39}{14}, -\frac{19}{7}$

d $-\frac{2}{5}, -\frac{6}{9}, -\frac{11}{45}$

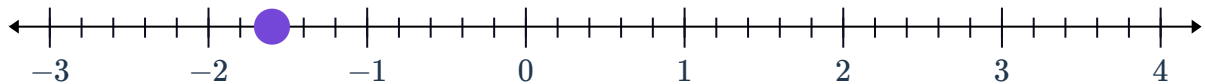
Arithmetic with directed fractions

 10 Use the plotted point on the following number lines to perform the indicated operation:

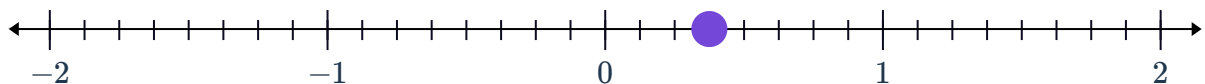
a $\frac{3}{5} + \frac{19}{5}$



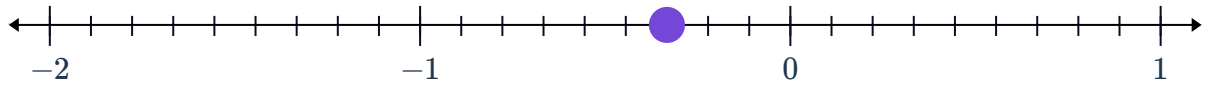
b $-\frac{8}{5} + \frac{19}{5}$



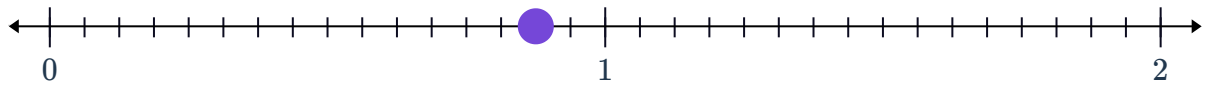
c $\frac{3}{8} - \frac{5}{8}$



d $\frac{5}{9} - \frac{8}{9}$



e $\frac{14}{16} - \frac{3}{16}$



11 Evaluate:

a $-\frac{7}{3} + \frac{17}{3}$

b $-\frac{26}{3} + \frac{16}{3}$

c $\frac{3}{5} - \frac{14}{5}$

d $\frac{5}{11} + \left(-\frac{9}{11}\right)$

e $-\frac{5}{9} + \left(-\frac{11}{9}\right)$

f $2\frac{2}{9} - \left(-\frac{5}{9}\right)$

g $-\frac{2}{9} - \left(-\frac{3}{5}\right)$

h $-\frac{2}{3} + \left(-\frac{1}{9}\right)$

Applications

12 If Patricia travels $4\frac{2}{5}$ km west, then $6\frac{3}{5}$ km east, find how far she is from her starting point.

13 This month, John spent $\frac{1}{4}$ of his allowance on food, $\frac{1}{8}$ on school projects, $\frac{3}{16}$ on transportation. What fraction of his allowance is left?

14 The average water level at a port is marked at 0 m, and a tide scale is used to measure the water level as it moves up and down with the tide.

When Jack begins his shift at 11 am, the tide scale reads $\frac{2}{5}$ m. Over the next 7 hours, the tide falls by $\frac{2}{3}$ m, and his shift ends. State the reading of the tide scale at the end of his shift.

- 15 The largest pumpkin in Shawn's farm weighs  $3\frac{3}{8}$ kg. This is $1\frac{5}{6}$ kg more than the average weight of pumpkins in his farm. Find the average weight of the pumpkins from Shawn's farm.
- 16 Willy and Bill are selling lemonade in the school fair. They have prepared $24\frac{3}{4}$ cups, which costed them \$18. Each serving of lemonade contains $1\frac{1}{4}$ cups, which they sell at \$2.
- a At the end of day, Willy and Bill checked the jug and there is still $3\frac{1}{2}$ cups left over. How many cups of lemonade did they sell that day?
- b Did Willy and Bill make a profit? Explain your answer.
- 17 Diane needs to purchase flour for a cooking class in school. The class is going to prepare a chocolate cake which requires $2\frac{1}{2}$ kg of flour, battered fish which requires $4\frac{3}{4}$ kg of flour, and banana bread which requires $3\frac{1}{4}$ kg of flour.

If the school already had 5 kg of flour, how many cups of flour Diane must buy?